

Zelvaritin Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-351/B

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. No changes to the approved analytical procedures are proposed in this submission.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Section 3.2.P.4 — Excipients

For the commercial formulation F-351/B, Explotab (Sodium starch glycolate, disintegrant) is used.

The approach described herein is consistent with USP-NF and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Formulation Development Rationale

The selection of Explotab is justified by lower moisture content mitigating hydrolytic degradation of the drug substance observed during development studies.

Current Commercial Specification

The current approved commercial specification (SPEC-EX-9580 Rev 05) lists Sodium starch glycolate as Primojel.

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. No changes to the approved analytical procedures are proposed in this submission. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Cleaning verification swab results were below the health-based exposure limits derived for the product. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

The control strategy links critical quality attributes to critical process parameters identified during development. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Cleaning verification swab results were below the health-based exposure limits derived for the product. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.